



Indian Institute of Technology Gandhinagar

MA 104: ODE

Semester-II, Academic Year: 2025-26

Tutorial – 2

T2.1. General Solution of Differential Equations

Find the general solution of the following differential equations:

(a) $2xy y' + (x - 1)y^2 = x^2 e^x$.

(b) $y' = 6(y - 2.5) \tanh(1.5x)$.

T2.2. Electric Field - Orthogonal Trajectories

The equipotential surfaces of an electric field between two concentric cylinders along the z -axis are given by $u(x, y) = x^2 + y^2 = c$, which represent circular cylinders in three-dimensional space. Find the family of orthogonal trajectories, i.e., the curves representing the electric field lines.

T2.3. Temperature Field - Heat Flow Lines

In a region occupying the upper half-plane $y > 0$, the isotherms (curves of constant temperature) are described by $4x^2 + 9y^2 = c$. Find the orthogonal trajectories, which correspond to the paths along which heat flows in a homogeneous medium without heat sources or sinks.

T2.4. Radioactive Decay Chain

Radium decays to radon, which in turn decays to polonium. If at $t = 0$ a sample is pure radium, determine how much radon it contains at time t . Let N_0 denote the number of radium atoms at $t = 0$, $N_1(t)$ the number of radium atoms at time t , and $N_2(t)$ the number of radon atoms at time t . Moreover, let λ_1 and λ_2 be the decay constants of radium and radon, respectively.

T2.5. Linear Differential Equation and Integrating Factor

Consider the linear differential equation $y' + p(x)y = r(x)$. Rewrite this equation in the form $M(x, y) dx + N(x, y) dy = 0$. Show that the resulting differential equation is not exact.

Find a suitable integrating factor and convert it into an exact differential equation. Finally, use the associated techniques to figure out the solutions $u(x, y) = c$.

T2.6. Show that for a linear ODE $y' + p(x)y = r(x)$, with continuous functions $p(x)$ and $r(x)$ in $|x - x_0| \leq a$, a Lipschitz condition holds.

T2.7. Find the orthogonal trajectories of $x^2 + y^2 - 2Cx = 0$.